

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons) — Semester VI Examination, 2013-14

Paper No. MTB-603: Numerical Analysis

Time: 3 Hours

Full Marks: 50

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks. Use of calculator is allowed.*

Question 1.

- (a) Using the Newton-Raphson method, derive the iterative formula $x_{n+1} = \frac{1}{3} \left(2x_n + \frac{N}{x_n^2} \right)$ to calculate the cube root of N . [2]
- (b) Find the value of $y(3)$ if $y(0) = 1$, $y(1) = 3$, $y(2) = 9$, $y(4) = 81$. [2]
- (c) Prove that $[\ln f(x)]' = \ln \left(1 + \frac{\Delta f(x)}{f(x)} \right)$. [2]
- (d) Define an ill-conditioned matrix. Give an example. [2]
- (e) What are the differences between Euler's method and the modified Euler's method for solving first-order initial value problems? [2]

Question 2.

- (a) A real root of $f(x) = x^3 - 2x - 5 = 0$ lies in $(2, 3)$. Perform four iterations of the Regula-Falsi (False Position) method to find this root, writing results to four decimal places. [5]
- (b) Obtain the order of convergence of the Secant method. [5]

Question 3.

- (a) Derive the Lagrange interpolation formula to approximate a given function by an n -th degree polynomial. [5]
- (b) From the following table, find x (correct to two decimal places) for which y is maximum, and also find the maximum value of y :

x	1.2	1.3	1.4	1.5
y	0.9320	0.9636	0.9855	0.9975

[5]

Question 4.

- (a) Obtain the error in Simpson's three-eighth rule of numerical integration. [5]
- (b) Evaluate $\int_0^1 \frac{1}{1+x^2} dx$ using Simpson's one-third rule with six equal sub-intervals (to four decimal places). [5]
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Question 5.

- (a) From the Taylor series for $y(x)$, find $y(0.1)$ correct to four decimal places, given that $\frac{dy}{dx} = x^2 - y^2$, $y(0) = 1$. [5]
- (b) Use the fourth-order Runge-Kutta method to solve $\frac{dy}{dx} = x + y^2$, $y(0) = 1$, with step size $h = 0.1$ on $[0, 0.2]$ to four decimal places. [5]
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Question 6.

- (a) Using the Gauss-Jacobi iterative method, solve the system:

$$x + 5y + 2z = -6, \quad 4x + y + z = 2, \quad x + 2y + 3z = -4$$

starting with initial approximation $[0.5, -0.5, -0.5]^T$, correct to three decimal places. [5]

- (b) Solve the following system using Gaussian elimination:

$$\begin{pmatrix} 2 & 1 & 1 \\ 4 & 0 & 2 \\ 3 & 2 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}$$

[5]

Question 7.

- (a) Describe the Jacobi rotation method to reduce a symmetric matrix to diagonal form.
[5]
- (b) Find the largest eigenvalue and corresponding eigenvector of:

$$A = \begin{pmatrix} 4 & 1 & 0 \\ 1 & 20 & 1 \\ 0 & 1 & 4 \end{pmatrix}$$

correct to three decimal places using the power method. [5]